ROAD_TRAFFIC_ACCIDENT_ANALYSIS

ROAD_TRAFFIC_ACCIDENT_ANALYSIS
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Scope: This process covers the academic road traffic accident analysis workflow from defining the study and collecting multi-country accident data through data quality preparation, Oracle Database implementation, SQL analysis, Power BI presentation, reporting, and road-safety recommendations. It begins with study definition and ends with documented findings and recommendations.

Diagram key
- ▭ Process
- ◆ Decision
- ⬭ Terminator

ROAD_TRAFFIC_ACCIDENT_ANALYSIS
Start
1. Define the Study<br>Objectives, questions, countries, scope
2. Collect Data<br>Collision, vehicle, casualty, road-safety datasets
3. Check Data Sources<br>Source, country, years, coverage, limitations
4. Clean &amp; Validate<br>Missing values, duplicates, formats, inconsistencies
Data issues resolved<br>or documented?
Yes
No: correct or document issues
5. Prepare Data<br>Standardize fields and categories; preserve originals
6. Design Database Entities, attributes, keys, relationships, ER diagram
7. Load into OracleTables, constraints, validated datasets
8. Analyze with SQLFrequency, causes, severity, trends, locations
9. Present Findings Power BI dashboards, reports, recommendations
End

Road Traffic Accident Analysis Workflow

Diagram information
This diagram presents an end-to-end process for analyzing road traffic accident data. It covers study definition, data collection and validation, data preparation, Oracle database design and loading, SQL analysis, and presentation of findings through Power BI dashboards and reports. Data-quality issues are corrected or documented before the workflow proceeds.

Standard Operating Procedure for Road Traffic Accident Analysis
1. Start the road traffic accident analysis process.
2. Define the study objectives and scope.
    2.1. Establish the research objectives and questions.
    2.2. Identify the countries or regions to be studied.
    2.3. Define the study period, population, and analytical scope.
3. Collect relevant datasets.
    3.1. Gather collision and crash records.
    3.2. Collect vehicle and casualty data.
    3.3. Obtain applicable road-safety and contextual datasets.
4. Check and document data sources.
    4.1. Record the source of each dataset.
    4.2. Verify the countries, years, and geographic coverage.
    4.3. Identify data limitations, gaps, and methodological differences.
5. Clean and validate the collected data.
    5.1. Identify and address missing values.
    5.2. Remove or resolve duplicate records.
    5.3. Standardize data formats and field values.
    5.4. Investigate and resolve inconsistencies across datasets.
6. Determine whether data issues have been resolved or documented.
    6.1. If issues are unresolved, correct them where possible or document the limitations, then repeat data cleaning and validation.
    6.2. If issues are resolved or adequately documented, proceed to data preparation.
7. Prepare the validated data for analysis.
    7.1. Standardize field names, categories, codes, and measurement units.
    7.2. Ensure compatible structures across datasets.
    7.3. Preserve the original source data separately from transformed data.
8. Design the database structure.
    8.1. Define entities, attributes, primary keys, and foreign keys.
    8.2. Establish relationships among collision, vehicle, casualty, location, and related entities.
    8.3. Create and review the entity-relationship diagram.
9. Load the prepared data into Oracle.
    9.1. Create the required tables and database objects.
    9.2. Apply constraints and relationship rules.
    9.3. Load the validated datasets.
    9.4. Verify record counts, relationships, and data integrity after loading.
10. Analyze the data using SQL.
    10.1. Calculate accident and casualty frequencies.
    10.2. Analyze contributing causes and severity levels.
    10.3. Identify trends over time.
    10.4. Examine geographic locations and accident concentrations.
11. Present the findings.
    11.1. Build Power BI dashboards and reports.
    11.2. Communicate significant patterns, risks, and trends.
    11.3. Provide evidence-based road-safety recommendations.
12. End the analysis workflow.